Three LLMs on Four Blackhole Chips

Aditya Sriram

Department of Computer Science, Stanford University adisram@stanford.edu

Project

End-to-end bringup of three modern LLM families on a single 1×4 **Blackhole P150** mesh: dense GQA, MoE with Gated DeltaNet recurrence, and a hybrid sliding/global-attention model. Each model walks from HF reference to a continuous-batching HTTP chat server, gated against a CPU fp32 oracle.

Hardware

Mesh	1×4 Blackhole P150 (4 chips)
DRAM	32 GB GDDR6/chip, **404 GB/s** stream
On-chip	110 Tensix cores, 1.39 MiB L1/core
Host link	1.4 GB/s tilize-bound

Models Brought Up

Model	Params	Architecture	Status
Qwen3.6-27B	27B	Dense GQA, TP4	shipped
Qwen3.6-35B-A3B	35B	MoE + Gated DeltaNet	$B{=}1$
Gemma 4 12B IT	12B	Hybrid sliding/global	shipped

Methodology

A staged ladder, each rung gated on cosine vs. a CPU fp32 oracle: single-layer → full forward → multi-step decode → long context → trace → continuous batching → HTTP. Each new model reused ~80–90% of the prior bringup. Validation is teacher-forced cosine *and* MAD per layer; six of seven silent-failure classes are invisible at decode pos 0 alone.

What Mattered

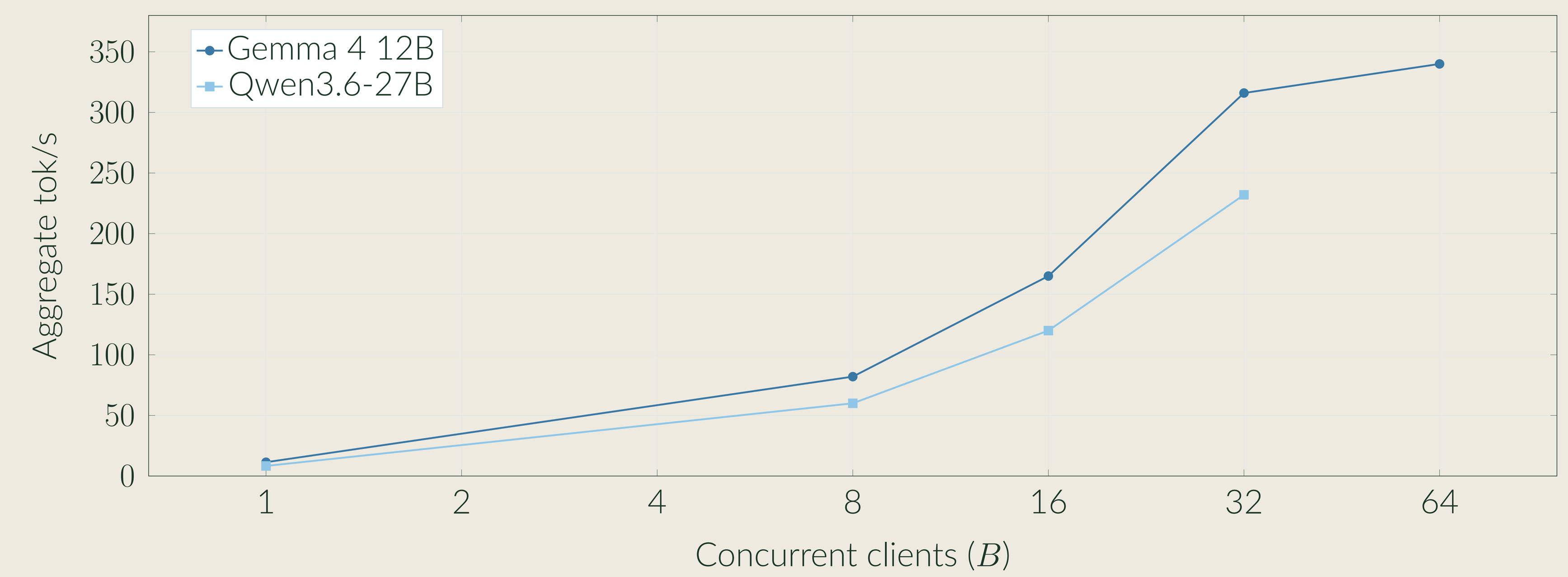
Three changes account for most of the throughput delta over a naive port.

- **Paged SDPA, HiFi2 fp16 accumulation** — kernel-time win; $+62\%$ on 27B TP single-stream.
- **Vocab-sharded lm_head, on-device argmax** — a 600 KB host readback collapses to 8 bytes; $+8\%$ Gemma 4, $+5\%$ 27B.
- **Two-phase trace warmup** — compile every path first, capture all traces back-to-back; otherwise decode JIT corrupts prefill trace memory.

Headline

Gemma 4 12B: 316 tok/s aggregate at 32 concurrent clients.
Qwen3.6-27B: 232 tok/s aggregate at 32 concurrent clients.
28–29× near-linear scaling from $B{=}1$ via continuous batching.

Throughput vs. Concurrent Clients



Diminishing returns past $B{=}32$ (316 → 340 at $B{=}64$).

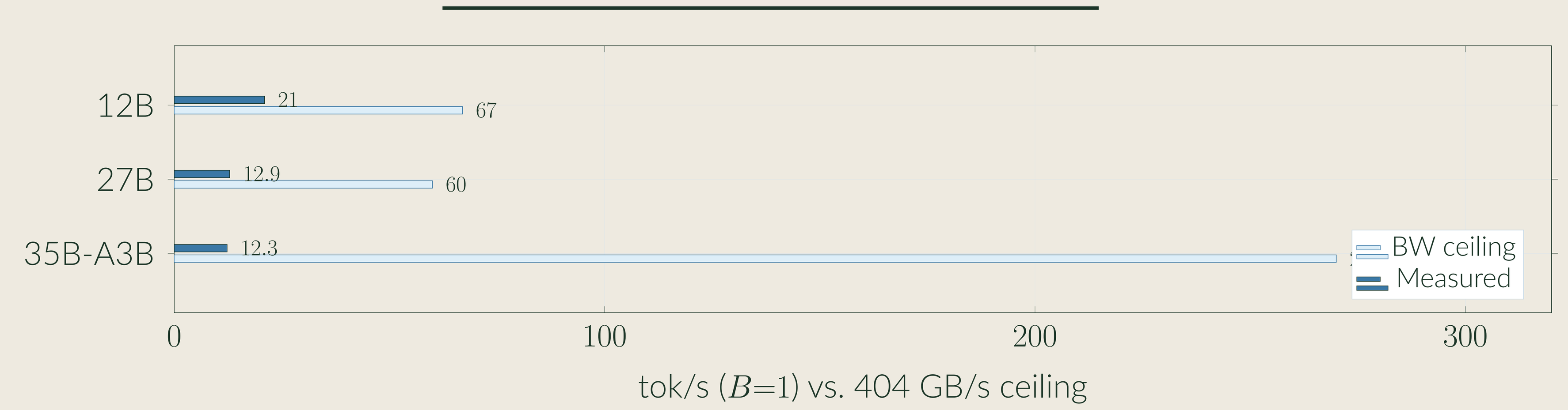
Single-Client Streaming (HTTP, $B{=}32$ traced)

Model	Prefill	Decode
Gemma 4 12B	**78 ms/tok** (TTFT 1.40 s)	**17.4 tok/s**
Qwen3.6-27B	**94 ms/tok** (TTFT 1.70 s)	**11.5 tok/s**

Decode is 83% of the $B{=}1$ dev-harness ceiling.

Roofline: Measured vs. BW Ceiling ($B{=}1$)

Model	ms/tok	tok/s	% ceiling
Gemma 4 12B	**47.5**	**21.0**	31%
Qwen3.6-27B (TP4)	**77.0**	**12.9**	22%
Qwen3.6-35B-A3B	**81.2**	**12.3**	4.6%



Correctness

Three orthogonal gates, run per model on every change.

Gate	27B	35B-A3B	Gemma 4
HF cosine ladder	PASS	$\cos_{L32} \geq 0.987$	PASS
Multi-turn KV retention	PASS	3/3 PASS	PASS
Traced = Eager	PASS	PASS	**100/100**
Needle haystack	$L{=}500$	coherent	3/3 ($L \leq 512$)

Validation is against a CPU fp32 oracle with per-sub-op hooks. Cosine alone is insufficient ($\cos(a, ka){=}1$); MAD is reported at every probed position.

Continuous Batching

Iteration-level Orca-style scheduler: per-step admission and eviction, per-slot paged SDPA over a shared KV pool, slot-level content-keyed prefix cache. Each slot advances at its own position in a single batched forward.

Prefix Cache in Practice

Two-turn 27B chat over HTTP:

Turn	Prompt	Prefill
1 (cold)	105 tok	19.26 s
2 (PC hit)	172 tok	**3.06 s**

Turn 2 is a **6.3**× prefill speedup despite a longer prompt: the cached KV is reused for the shared prefix and only the suffix is computed.

Conclusion

A single 1×4 Blackhole P150 mesh, with a disciplined bringup recipe, takes a modern LLM from spec to byte-stable HTTP chat at **316 tok/s** on 12B and **232 tok/s** on 27B. The open stack lets a small team bring up new architectures (MoE, hybrid attention) without a vendor roadmap.